(ii)
Speed and RAM

海量数据的计算和挖掘成为主要瓶颈

Biggest hurdles	#cited
Data management/informatics/storage	19
General cost	5
Sample prep/workflow	4
DNA sample access	4
Funding	3
Quality and reproducibility	2
Regulatory/legal constraints	2
Capacity/access to more sequencers	2
Others	1

Source: J.P. Morgan survey

If we simply run BLASTX on Amazon EC2



- 95GB == 195,600 node hours (on Nehalem 8core, 16GB)
- illumina Hiseq2000 = 2x100GB/run
- cost is purely BLAST, no storage or transfer cost
- values are in Amazon EC2 (*wilkening#et#al,#EEE#Cluster09*)

Computation turns out to be …

CPU-intensive jobs
(read mapping, metagenomics)

10 454 runs

BLAST NR



800 CPUs

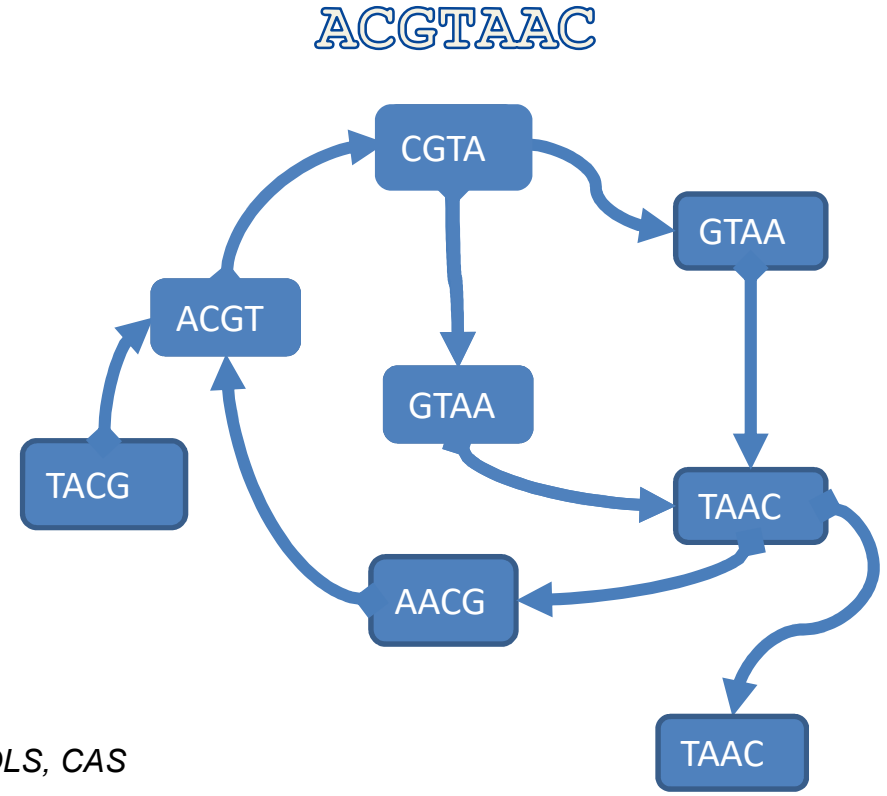
~300 hours



2 CPUs

~14 years

RAM-intensive jobs
(genome assembly)



by F Zhao @BIOLS, CAS

Performance in Amazon EC2

! " #$%4, ' 5164 7,)(, 82 19, 231)- 1:%2, ((4, ' /

	Asian Individual Genome		
Data Loading	3.3 B reads	106.5 GB	$10.65
Data Transfer	1h :15m	40 cores	$3.40
Setup	0h : 15m	320 cores	$13.94
Alignment	1h : 30m	320 cores	$41.82
Variant Calling	1h : 00m	320 cores	$27.88
End-to-end	4h : 00m		$97.69

Analyze an entire human genome for ~$100 in an afternoon.
Accuracy validated at >99%

Searching for SNPs with Cloud Computing.
Langmead B, Schatz MC, Lin J, Pop M, Salzberg SL (2009) *Genome Biology.* **10**:R134!